

AMD DEVELOPER HACKATHON — ACT II · TRACK 1

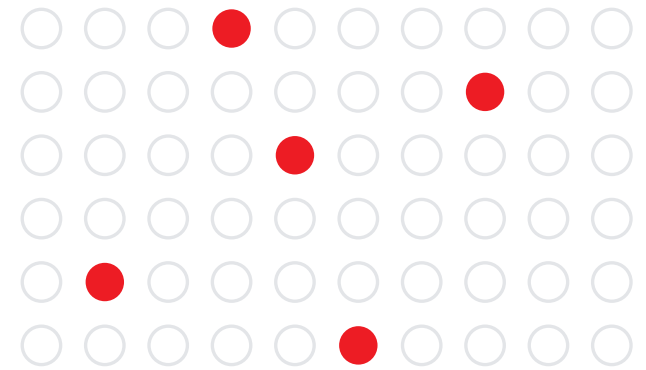
FrugalRouter

A hybrid routing agent that answers 19 varied tasks with the fewest external tokens possible.

Local-first inference · deterministic verification · budgeted escalation

`ghcr.io/manan-tech/frugalrouter:v3`

`github.com/manan-tech/frugalrouter`



the token budget — barely touched

The problem



AI spend control — without a quality tax

Enterprises want to cut LLM inference cost, but every saved token is worthless if accuracy drops below the bar.



One agent, eight task types

Track 1: answer 19 varied tasks — math, code generation & debugging, logic, NER, sentiment, summarisation, factual Q&A — above an accuracy gate.



Fireworks tokens are the score

Every external token counts against you. The cheapest correct answer wins.

THE GRADING BOX

2

vCPUs

4 GB

RAM

10 min

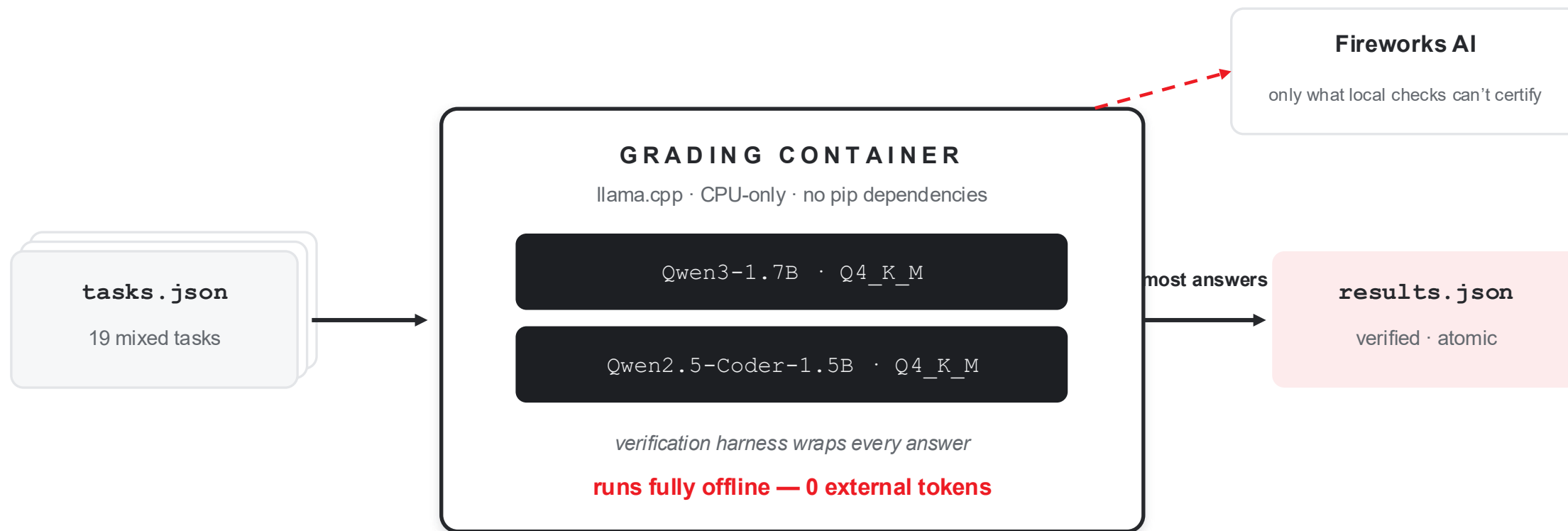
wall clock

0

GPUs

linux/amd64 container · offline except the Fireworks API · exit 0 in time, or score nothing

Don't route to the cheapest cloud model. Make cloud calls unnecessary.



Two small quantized LLMs are baked into the 2.24 GB image and run entirely on the grading CPUs. Local inference costs zero external tokens; Fireworks is reserved for the few answers local verification can't certify.

Verification, not trust

Small models can't be trusted raw — nothing ships unverified.



Math

The model writes a Python program. We execute it, then majority-vote the computed number across samples.



Code

Two independently sampled implementations must agree on observed behavior — disagreement triggers a third and a vote.



Logic

Entities and constraints extracted to JSON, every assignment brute-forced — an answer ships only with a proven unique solution.



Formats

Grammar-constrained decoding (GBNF) makes invalid labels impossible; sentence and word counts are enforced programmatically.

verification outcome



confidence score



escalate only if conf < 0.55

Smart escalation



Weakest answers, cheapest models first

Below-threshold answers escalate sequentially, cheapest-estimated-first, with reasoning_effort: low and tight output caps.



Live-measured most frugal model

We benchmarked every serverless model: gpt-oss-120b at low reasoning effort — 137 tokens for a complete word problem.



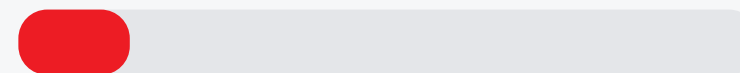
Factual always escalates

Sample agreement can't verify facts — and factual is the cheapest category to fix.

HARD GLOBAL BUDGET

900 tokens

cap across the entire 19-task run



one escalated task $\approx$ 110–320 tokens

ESCALATION_BUDGET_TOKENS = 0

→ fully local. Zero API calls, still finishes every task.

Engineering for a hostile clock



Speed governor

A startup probe measures real tok/s and sets sampling depth: full → lean → panic. Slow CPUs degrade gracefully, never fatally.



Atomic output

results.json is valid JSON after every single task, and each task runs in its own exception boundary.



Guaranteed clean exit

The watchdog forces a flush at 8:25 — the container structurally cannot time out or emit invalid output.

Results

40-task unseen-variant suite · LLM-judged · at grading constraints (2 vCPU / 4 GB, native x86)

90.0%

accuracy
zero escalation

95.0%

accuracy
with escalation

**~300–
700**

expected tokens
full 19-task set

232 s

19 tasks, wall clock
measured

llama.cpp

Qwen3-1.7B

Qwen2.5-Coder-1.5B

Fireworks AI

Docker

GitHub Actions CI

2.24 GB self-contained image · pure-stdlib Python orchestrator · MIT license

FrugalRouter — verified answers, almost free.